Lctr 1: Analog Signals in Both Time and Frequency Domains

Jianhua Liu

Fall 2026

Contents

1.1 Introduction . . . 2
1.2 Review of basic functions . . . 2
- 1.2.1 Definitions of cosine and sine functions on triangles . . . 2
- 1.2.2 Properties of cosine and sine functions . . . 2

1.3 Review of Euler's formula . . . 4
- 1.3.1 Definition of complex number . . . 4
- 1.3.2 Euler's formula . . . 4

1.4 Basics of sinusoidal signals . . . 4
- 1.4.1 Single-frequency complex sinusoid . . . 4
- 1.4.2 Single frequency real sinusoid . . . 5
- 1.4.3 Expressing a real sinusoid using complex sinusoids . . . 5
- 1.4.4 Spectrum of a real sinusoid . . . 5
- 1.4.5 Why using complex sinusoids? . . . 5
- 1.4.6 Superposition of harmonic signals . . . 7
- 1.4.7 Spectrum of superposition of harmonic signals . . . 7

1.5 Periodic signals . . . 7
- 1.5.1 Definition of periodic signals in time domain . . . 7
- 1.5.2 Fourier series of periodic signals . . . 8
- 1.5.3 Fourier series coefficients to spectrum . . . 8

1.6 Fourier transform . . . 8
- 1.6.1 Extending the period to infinity . . . 8
- 1.6.2 Definition of Fourier transform . . . 8

1.7 Dirac delta function . . . 8
- 1.7.1 Mathematical definition of Dirac delta function . . . 8
- 1.7.2 The sampling property of the Dirac delta . . . 9
- 1.7.3 Extending the Fourier transform of sinusoid with Dirac delta . . . 9
- 1.7.4 A train of Dirac delta function and its Fourier series . . . 9

1.8 Sampling theorem . . . 10
- 1.8.1 Sampling using the Dirac delta . . . 10
- 1.8.2 Choosing sample rate to avoid spectrum overlap . . . 10
- 1.8.3 Signal reconstruction . . . 10

1.9 Discrete Fourier analysis . . . 10
1.9.1 DTFT (Discrete Time Fourier Transform) for discrete time signals . . . 10
1.9.2 DFS (Discrete Fourier Series) . . . 11
1.9.3 DFT (Discrete Fourier Transform) . . . 12
1.9.4 DFT vs FFT . . . 12

1.1 Introduction

The digital signal processing (DSP) class deals with discrete signals, which can be generated by sampling analog signals.

In this lecture, we review sinusoidal signals and then discuss general analog signals. We also discuss briefly sampling theorem, which is the foundation to create discrete signals from analog signals.

Specifically, we discuss analog signals in both time- and frequency domains, including the following:

- Waveform in time domain.
- Spectrum in frequency domain.
- Sampling theorem.
- Principle of Discrete Fourier Transform.

1.2 Review of basic functions

1.2.1 Definitions of cosine and sine functions on triangles

The definition of cosine and sine functions can be shown easily using a triangle, such as the upper one in Figure 1:

$$\cos\theta = x/r,$$
$$\sin\theta = y/r.$$

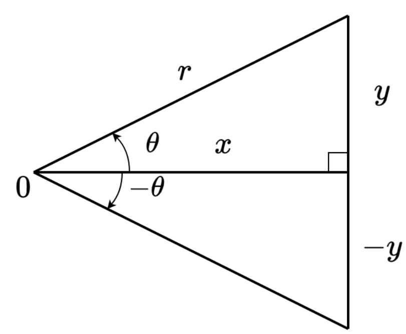

Figure 1: Mirror triangles for the definition and illustration of cosine and sine functions.

1.2.2 Properties of cosine and sine functions

From the bottom triangle of Figure 1 we can see:

$$\cos(-\theta) = \cos\theta,$$
$$\sin(-\theta) = -\sin\theta.$$

We can also see the following from Figure 2, where the above r is changed to 1, that

$$\cos(\frac{\pi}{2} - \theta) = \sin\theta,$$
$$\sin(\frac{\pi}{2} - \theta) = \cos\theta,$$
$$\cos(\pi + \theta) = -\cos\theta,$$
$$\sin(\pi + \theta) = -\sin\theta.$$

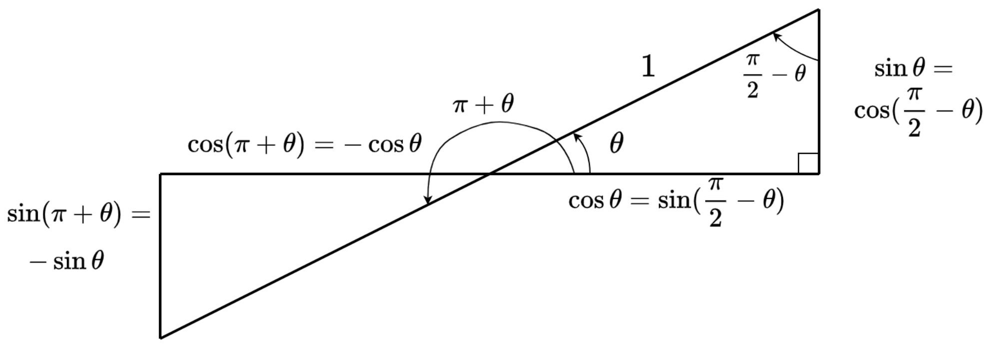

Figure 2: Opposite triangles for the illustration of cosine and sine properties.

Using the above properties, we have the following properties:

$$\cos(\theta - \frac{\pi}{2}) = \cos(\frac{\pi}{2} - \theta) = \sin\theta,$$
$$\sin(\theta - \frac{\pi}{2}) = -\sin(\frac{\pi}{2} - \theta) = -\cos\theta,$$
$$\cos(\frac{\pi}{2} + \theta) = \cos(\frac{\pi}{2} - (-\theta)) = \sin(-\theta) = -\sin\theta,$$
$$\sin(\frac{\pi}{2} + \theta) = \sin(\frac{\pi}{2} - (-\theta)) = \cos(-\theta) = \cos\theta,$$
$$\cos(\pi - \theta) = \cos(\pi + (-\theta)) = -\cos(-\theta) = -\cos\theta,$$
$$\sin(\pi - \theta) = \sin(\pi + (-\theta)) = -\sin(-\theta) = \sin\theta.$$

Additionally, we have, for integer k:

$$\cos(\theta + 2k\pi) = \cos\theta,$$
$$\sin(\theta + 2k\pi) = \sin\theta.$$

To see the relationship between the rotation on a circle and a sine function, see https://www.mathsisfun.com/algebra/trig-sin-cos-tan-graphs.html.

1.3 Review of Euler's formula

1.3.1 Definition of complex number

The complex number can often be expressed as i or j. Here we use j to be consistent. It is defined as $j = \sqrt{-1}$.

1.3.2 Euler's formula

For a given angle ϕ, we have

$$e^{j\phi} = \cos\phi + j\sin\phi,$$

where $e \approx 2.71828$ is the base for the natural logarithm. This is often illustrated using a unit circle, as shown in Figure 3.

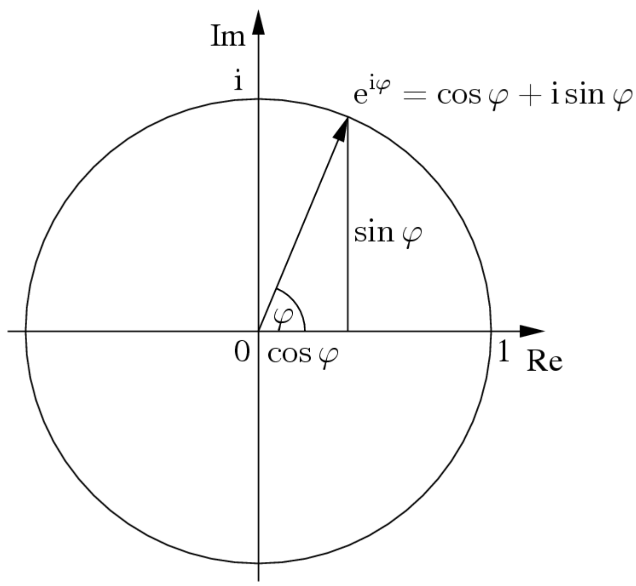

Figure 3: Unit circle for expressing the complex sinusoids in terms of cosine and sine functions.

The conjugate of the above $e^{j\phi}$ is

$$(e^{j\phi})^* = e^{-j\phi} = \cos(-\phi) + j\sin(-\phi) = \cos\phi - j\sin\phi.$$

To see the relationship between the rotation on a circle and a sine function, see https://www.mathsisfun.com/algebra/trig-sin-cos-tan-graphs.html.

1.4 Basics of sinusoidal signals

1.4.1 Single-frequency complex sinusoid

A single frequency complex sinusoid, a function of time t has the following general form:

$$x_C(t) = Ae^{j(2\pi Ft+\theta)},$$

with
whit the following parameters:

- A, the amplitude of the signal, the radius of the circle if we draw $x_C(t)$ on a complex plane.
- F, the (ordinary) frequency in Hz. Often, we can use $\Omega = 2\pi F$, the **angular** frequency, to simplify the notation.
- θ, the initial phase, the phase of the function when t is 0.

The above parameters are usually classified into two groups:

- $X_C = Ae^{j\theta}$, the complex amplitude, specifies the amplitude and initial phase.
- F, the frequency, determines how fast the complex sinusoidal signal changes.

1.4.2 Single frequency real sinusoid

If we only take the real part of the above complex sinusoid, we will have the following single-frequency real sinusoidal function:

$$x(t) = A\cos(2\pi Ft + \theta).$$

See the top part of Figure 4 for an illustration of the waveform of a sinusoidal function. The bottom part shows its spectrum, which is defined below.

1.4.3 Expressing a real sinusoid using complex sinusoids

The real sinusoidal signal $x(t)$ can be expressed using $x_C(t)$ as shown below:

$$\begin{aligned} x(t) &= \Re x_C(t), \\ &= \frac{x_C(t) + x_C^*(t)}{2}, \\ &= \frac{A}{2}e^{j\theta}e^{j2\pi Ft} + \frac{A}{2}e^{-j\theta}e^{-j2\pi Ft}. \end{aligned}$$

1.4.4 Spectrum of a real sinusoid

Given the values of A, θ, and F, the entire waveform of $x(t)$ can be completely determined. To express these three parameters in an effective way, especially when we consider the summation of sinusoids with multiple different frequencies, we can express them in a single graph, as shown in the bottom part of Figure 4.

1.4.5 Why using complex sinusoids?

It seems that we are making the matters more complicated by using complex sinusoids to express a simple real sinusoid. The reason, which will be illustrated in the class, is that in linear systems and analysis, we need to do *linear combinations* of multiple real sinusoids with the same frequency but different amplitudes and phases. The result of the linear combination is a sinusoid with the same frequency but unknow amplitude and phase. When expressing each real sinusoid using its complex counterpart, the calculation of the amplitude and phase is very straight using complex numbers.
straightforward

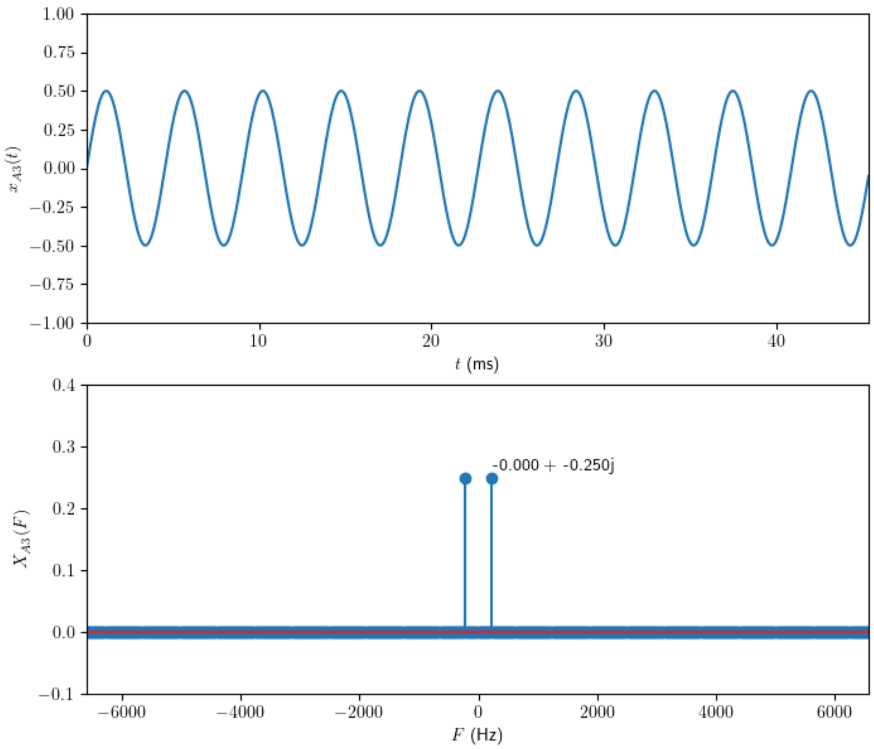

Figure 4: Waveform and spectrum of a sinusoid.

1.4.6 Superposition of harmonic signals

We can generate a signal as the summation of harmonics, as shown in the top of Figure 5.

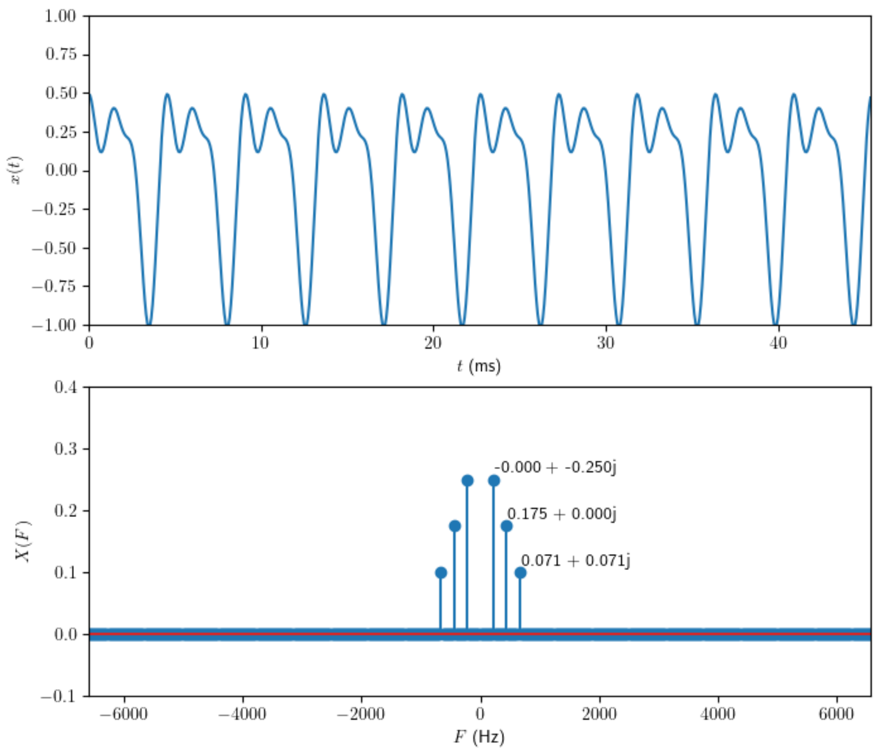

Figure 5: Waveform and spectrum of the summation of harmonics.

1.4.7 Spectrum of superposition of harmonic signals

The spectrum of superposition of harmonic signals really shows the advantage of spectrum—it include all the information we need to determine the time-domain signals.

1.5 Periodic signals

1.5.1 Definition of periodic signals in time domain

A signal $x(t)$ is said to be periodic with period P if $x(t + P) = x(t)$.

A signal is a continuous-time (CT) signal if for every t, the signal has a valid definition.

1.5.2 Fourier series of periodic signals

To stress that a signal is periodic, let us use $\tilde{x}(t)$ to express it. Then we have the following CTFS (continuous time Fourier series) pair:

$$X_k = \frac{1}{P}\int_P \tilde{x}(t)e^{-jk\Omega_0 t}dt, \quad \forall k \in \mathcal{I},$$
$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\Omega_0 t},$$

where $\int_P(\cdot)dt$ stands for integration over one period of the signal in TD and $\Omega_0 = 2\pi/P$ is the fundamental angular frequency of the periodic signal.

Note that **for CTFS, the TD waveform is periodic, and the FD spectrum is discrete, which suggests that periodicity in one domain implies discreteness in the other**.

1.5.3 Fourier series coefficients to spectrum

Values of the Fourier series coefficients constitute the spectrum, as shown in the bottom ~~as shown in the top~~ of Figure 5.

Will continue from here on 8/28/26.

1.6 Fourier transform

1.6.1 Extending the period to infinity

When we extend P of a periodic signal to infinity, the signal is not periodic any more, and the Fourier series will be modified to be Fourier transform.

1.6.2 Definition of Fourier transform

Fourier transform (FT) is only defined for non-periodic CT signal. Here is the definition.

Let $x(t)$ be a CT signal. Then we have the following CTFT pair (analysis/synthesis):

$$X(\Omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt,$$
$$x(t) = \mathcal{F}^{-1}\{X(\Omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)e^{j\Omega t}d\Omega,$$

where Ω is the angular frequency of a continuous time signal in radians per second. As we have discussed before, $\Omega = 2\pi F$.

Note that **for CTFT, both the TD waveform and the FD spectrum are continuous**.

1.7 Dirac delta function

Dirac delta function is a special virtual function defined for creating self-contained theory for signal analysis.

1.7.1 Mathematical definition of Dirac delta function

Dirac delta function is defined by the following conditions:

$$\delta(t) = \begin{cases} 0, & t \neq 0, \\ \infty, & t = 0. \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t)dt = 1.$$

~~###~~ Dirac delta function as the limit of function series

See class demo.

1.7.2 The sampling property of the Dirac delta

$$\int_{-\infty}^{\infty} \delta(t-\tau)s(t)dt = s(\tau).$$

1.7.3 Extending the Fourier transform of sinusoid with Dirac delta

We have discussed that we have the following expression

$$x(t) = \frac{A}{2}e^{j\theta}e^{j2\pi F_{o}t} + \frac{A}{2}e^{-j\theta}e^{-j2\pi F_{o}t}.$$

There is no way for us to do Fourier transform (FT) on this signal directly as the integration does not converge.

Yet, we can conjecture that the FT of a complex sinusoid is somehow related to a Dirac delta. Then, we can use the formula of inverse FT to see if the TD signal is indeed a complex sinusoid.

So, we conjecture that for $x_{\mathrm{C},F_{o}}(t) = e^{j2\pi F_{o}t} = e^{j\Omega_0 t}$, its FT is $X_{\mathrm{C},F_{o}}(\Omega) = 2\pi\delta(\Omega - \Omega_0)$. Then,

$$\begin{aligned} x_{\text{conjecture}}(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)e^{j\Omega t}d\Omega, \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\delta(\Omega-\Omega_0)e^{j\Omega t}d\Omega, \\ &= e^{j\Omega_0 t}. \end{aligned}$$

So, we know that the FT of $x(t) = \frac{A}{2}e^{j\theta}e^{j2\pi Ft} + \frac{A}{2}e^{-j\theta}e^{-j2\pi Ft}$ is

$$X(\Omega) = A\pi e^{j\theta}\delta(\Omega - \Omega_0) + A\pi e^{-j\theta}\delta(\Omega + \Omega_0).$$

1.7.4 A train of Dirac delta function and its Fourier series

Let's define the following train of pulse:

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT).$$

Then, by using CTFS, we have its Fourier series coefficients as

$$\Delta_k = \frac{1}{T}, \quad \forall k \in \mathcal{I},$$

and hence, the CTFT of $\delta_T(t)$ is

$$\Delta(\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_S),$$

where $\Omega_S = 2\pi/T$.

Here, T is the sample interval, and $F_S = 1/T$ is the sample rate, which is also called sample frequency when we do Fourier analysis.

1.8 Sampling theorem

1.8.1 Sampling using the Dirac delta

Let $x(t)$ be a CT signal and

$$x_S(t) = x(t) \cdot \delta_T(t)$$

be the CT *pulse*-sampled signal. Then, according to the property of CTFT, not proved here but can be done easily, we have

$$X_S(\Omega) = X(\Omega) * \Delta(\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_S).$$

Note that the above $*$ notation stands for convolution.

When $x(t)$ is bandlimited to $\Omega_S/2$, i.e.,

$$|X(\Omega)| = 0, \quad \forall |\Omega| > \pi/T = \Omega_S/2,$$

there will be no aliasing and we can recover $X(\Omega)$ from $X_S(\Omega)$. This leads to the following sampling theorem.

1.8.2 Choosing sample rate to avoid spectrum overlap

The relationship between the sample rate and the spectrum of the sampled signal is shown in Figure 6. We can see that when the sample rate is too low, we will have overlap in the spectrum, and this will destroy the spectrum, which leads to corrupted sampled signal.

the chance to recover the original

1.8.3 Signal reconstruction

that

We don't care much about the reconstruction of the DT signals to CT. See class illustrations for this topic.

1.9 Discrete Fourier analysis

1.9.1 DTFT (Discrete Time Fourier Transform) for discrete time signals

DTFT is the counterpart of CTFT for discrete time (DT) signals.

Let $x[n]$ be a DT sequence. Then we have the following DTFT pair:

$$X(e^{j\omega}) = \mathcal{F}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n},$$

$$x[n] = \mathcal{F}^{-1}\{X(e^{j\omega})\} = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega,$$

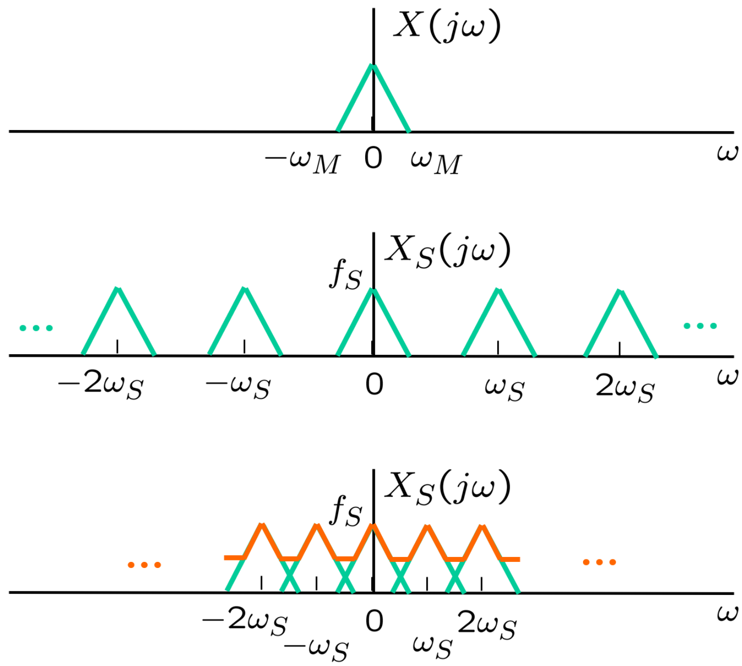

Figure 6: Spectra of the CT and sampled signals. Note that the notations in the text and the figure are not consistent. Here ω is the same as Ω in the text.

where ω is the angular frequency of discrete time signal in radians per sample.

Note that **for DTFT, the DT sequence is discrete, and the FD spectrum is periodic, which verifies the above suggestion that periodicity in one domain implies discreteness in the other**.

1.9.2 DFS (Discrete Fourier Series)

(If we had followed the same line of naming as demonstrated in CTFT ↔ CTFS, we would have named DFS as DTFS; but DFS ends up as the preferred term due to historical reasons.)

Let $\tilde{x}[n]$ be a periodic DT sequence with period N. Then we have the following DFS pair:

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}, \quad \forall k \in \mathcal{I},$$

$$\tilde{x}[n] = \frac{1}{N}\sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi}{N}kn}, \quad \forall n \in \mathcal{I}.$$

Note that $\tilde{X}[k]$ is also periodic with period N.

Note that **for DFS, both the DT sequence and the FD spectrum are discrete and periodic, which further verifies the suggestion that periodicity in one domain implies discreteness in the other**.

Due to the periodicity in both TD and FD, for simplicity of computation, we only need to consider one period in each domain. In this case, we call the transform DFT (Discrete Fourier Transform), as detailed in the next section.

1.9.3 DFT (Discrete Fourier Transform)

A DFT pair is defined as

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, \quad 0 \le k < N,$$
$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, \quad 0 \le n < N.$$

We can go between a periodic sequence (with period N) and its one period back and forth as

$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n-kN], \quad n \in \mathcal{I}$$
$$x[n] = \begin{cases} \tilde{x}[n], & 0 \le n < N \\ 0, & \text{otherwise} \end{cases}$$

DFT is often implemented by FFT (Fast Fourier Transform).

1.9.4 DFT vs FFT

Mathematically, there is not difference between DFT and FFT. The latter is just faster.

The choice of N is usually a power of 2 as this can lead to the best computational efficiency. Other numbers work as well.